

Topology-Aware Inverse Preference Learning for Robust Hybrid Voting Systems

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Abstract. Hybrid voting systems combine expert scores with crowd support, yet typically expose only the resulting elimination, not the crowd votes themselves. We ask what this censored feedback can and cannot reveal about hidden crowd preferences. Our method, Topology-Aware Inverse Preference Learning (TIPL), combines relational features, a utility anchor fit through the known aggregation rule, and an outcome-conditioned projection onto the set of feasible vote shares. We deliberately separate three questions that are often conflated: predicting future eliminations, checking consistency with a past outcome, and recovering the true votes. In controlled experiments with known ground-truth votes, rule-aware learning lowers total-variation error relative to a survival-only objective, while the projection step mainly restores feasibility rather than improving accuracy. A reproducible, chronologically ordered evaluation on a 34-season competition dataset finds no clear predictive edge over simple baselines. The resulting feasible vote sets remain wide, and our topological diagnostics find no evidence of persistent preference cycles in the rounds studied. Counterfactual stress tests further reveal a trade-off between expert alignment and crowd retention, rather than a uniform fairness gain. Taken together, these results position TIPL as a tool for assumption-dependent reconstruction and mechanism auditing: consistency with an observed elimination is not sufficient to identify the true crowd preferences behind it.

Keywords: Inverse preference learning · Hybrid voting · Partial identification · Topological diagnostics · Mechanism auditing

1 Introduction

Hybrid evaluation systems blend professional judgment with public participation, as seen in televised competitions, community rankings, and human-evaluation platforms. Expert assessments and audience support need not agree: technical performance, familiarity, and social influence can each pull collective outcomes in different directions [1,2]. When the audience’s vote totals are never revealed, the

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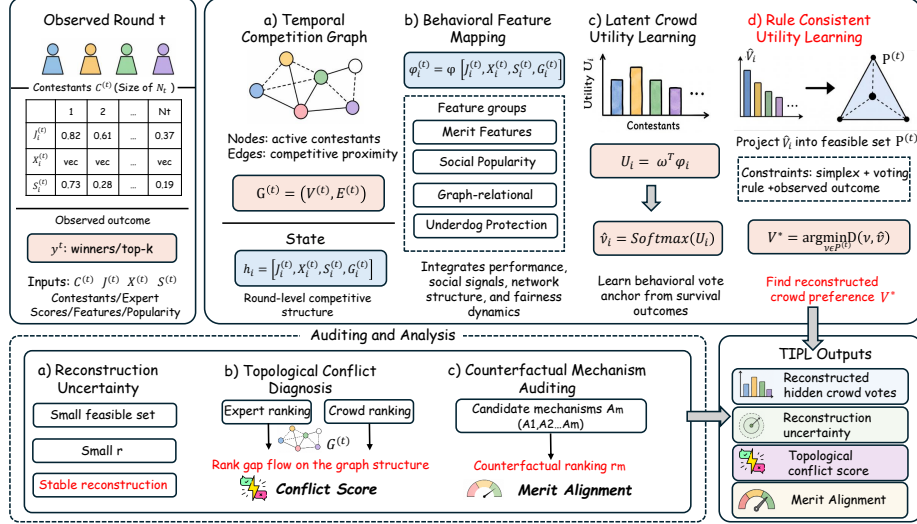


Fig. 1. Conceptual overview of TIPL. Round-level observations first produce a behavioral anchor; the observed outcome then enters a separate reconstruction stage. Operational constraints, information timing, and the distinct diagnostic quantities are defined in Sections 2–3.4. The popularity channel shown in the diagram is omitted from the chronological experiment because no contemporaneous measurements of it are available.

recorded elimination is only a censored glimpse of this combined process. Even so, characterizing which crowd preferences are *consistent* with an observed outcome is valuable for auditing a voting mechanism, even when those preferences cannot be pinned down exactly.

A central distinction runs through this paper: explaining an outcome is not the same as identifying the preferences that produced it. A distribution can reproduce every recorded elimination and still sit far from the actual votes. Likewise, a model trained purely to predict survival can end up encoding expert-score effects rather than genuine crowd preferences. These are two separate difficulties, not one: censoring makes the inverse problem underdetermined, while ignoring the known aggregation rule confounds the interpretation of any learned utility. High retrospective consistency, by itself, resolves neither.

TIPL addresses this by separating a learned behavioral anchor from the constraints imposed by the mechanism itself. Observable contestant features and a competition graph parameterize a vote-share model, and a likelihood built from the hybrid score incorporates the known expert scores when fitting historical eliminations. Once an outcome is observed, a constrained projection selects the distribution nearest the anchor that remains compatible with that outcome. The resulting feasible set, together with graph- and topology-based diagnostics, then supports a conditional form of mechanism analysis. Figure 1 summarizes this pipeline.

Our contribution is a reproducible formulation of this distinction together with a careful empirical test of it. First, we specify the feasible constraints and

the intrinsic geometry of the vote simplex they define. Second, we separate outcome-blind prediction from outcome-conditioned reconstruction, and evaluate actual vote recovery in a setting where ground truth is available. Third, we define explicit diagnostic and counterfactual quantities, including negative controls and a crowd-retention metric. We keep our empirical claims deliberately modest: the evidence supports TIPL as a useful auditing construction and confirms synthetic recovery under its stated assumptions, but it does not establish accurate real-world vote recovery, a predictive benefit from graph features, or a universally superior voting rule.

Related work. Revealed preference and inverse reinforcement learning both infer latent objectives from observed behavior [3,4,5]. Preference-based reinforcement learning typically relies on explicit pairwise comparisons [6], and recent inverse-preference-learning methods remove the need for a separate reward model in offline RL [7]; direct preference optimization exploits a similarly structured relationship between a policy and its implicit reward [8]. Our setting differs in an important way: an elimination combines a known expert signal with unobserved crowd votes, and we do not assume access to pairwise crowd labels or trajectories.

Learning-to-rank and computational social choice study the prediction and aggregation of preferences [9,2,10], but in our setting aggregation is itself part of the observation process rather than a downstream decision applied afterward. Graph representations are a natural source of contextual features [11,12], and combinatorial Hodge theory offers a principled way to separate a pairwise flow from its best additive explanation [13]. TIPL draws on graph statistics and explicitly defined diagnostics rather than a learned graph neural network; these are established tools, and combining them does not by itself yield an identifiability guarantee.

2 Observations and Identification

In round t , let C_t contain n_t active contestants, let $q_t \in \Delta^{n_t-1}$ denote normalized expert scores, and let $v_t \in \Delta^{n_t-1}$ denote the hidden crowd vote shares, where the simplex is $\Delta^{n-1} = \{v \in \mathbb{R}^n : v \geq 0, \mathbf{1}^\top v = 1\}$. Observable features are X_t , and $E_t \subset C_t$ records the eliminated contestants. Under percentage aggregation with a known expert weight $\alpha_t \in (0, 1)$,

$$c_{t,i} = \alpha_t q_{t,i} + (1 - \alpha_t) v_{t,i}. \quad (1)$$

Contestants with the lowest combined scores are eliminated. Our main real-data experiment restricts attention to single-elimination rounds with $\alpha_t = 1/2$; rank aggregation and a judge’s discretionary save induce different observation maps, and we do not fold them silently into this linear rule.

A feasible explanation is not an identified preference. Consider equal expert scores with the third contestant eliminated. Both $v = (0.40, 0.35, 0.25)$ and $v' = (0.70, 0.20, 0.10)$ are compatible with this outcome, yet their total-variation

Table 1. Three evaluation questions require different information. An observed test outcome may be used to score a prediction, but it cannot also be an input to that prediction.

Task	Information supplied	Evaluation target
Prediction	q_t, X_t and past outcomes	Future elimination E_t
Reconstruction	q_t, X_t, E_t and the rule	Constraint satisfaction
Vote recovery	Same reconstruction inputs	Hidden v_t ; synthetic data only

distance is 0.30. If v' happens to be the anchor, projection leaves it unchanged even when v was the true generating vote. Consistency with the observed outcome therefore cannot, on its own, bound vote-share error. A parametric utility model does let us share information across rounds, but its interpretation still hinges on the model, the chosen features, and the correctness of the assumed observation rule. Table 1 fixes the terminology we use throughout.

3 Method

3.1 Relational Features and Rule-Aware Utility Learning

Let $m_{t,i}$ denote the observed mean judge score divided by ten. We treat the round graph as complete, with affinities

$$a_{t,ij} = \exp\left[-\frac{(m_{t,i} - m_{t,j})^2}{2\sigma_g^2}\right], \quad i \neq j, \quad a_{t,ii} = 0. \quad (2)$$

From this graph we extract weighted PageRank, weighted closeness, and weighted clustering; closeness is computed using reciprocal affinity as an edge length, since unweighted closeness would be constant on a complete graph. The remaining observed features are the technical score, a partner’s average score in strictly earlier seasons, a sigmoid-transformed underdog feature, and the change from the contestant’s most recently observed score. We treat these as predictive descriptors rather than measurements of any underlying psychological cause. Section 4.1 gives the full construction and preprocessing details.

A linear utility produces the anchor

$$U_{t,i} = w^\top \phi_{t,i}, \quad \hat{v}_{t,i}(w) = \frac{\exp U_{t,i}}{\sum_{j \in C_t} \exp U_{t,j}}. \quad (3)$$

For a single eliminated contestant e_t , we learn through the combined score rather than treating survival probabilities as votes directly:

$$\begin{aligned} \hat{c}_{t,i}(w) &= \alpha_t q_{t,i} + (1 - \alpha_t) \hat{v}_{t,i}(w), \\ p_w(e_t = i \mid q_t, X_t) &= \frac{\exp[-n_t \hat{c}_{t,i}(w)/\tau]}{\sum_j \exp[-n_t \hat{c}_{t,j}(w)/\tau]}, \end{aligned} \quad (4)$$

$$w^* = \arg \min_w \left\{ -\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \log p_w(e_t \mid q_t, X_t) + \lambda \|w\|_2^2 \right\}. \quad (5)$$

Algorithm 1 TIPL with an explicit information boundary**Require:** Historical training rounds; current expert scores q_t and features X_t

- 1: Select λ on an earlier validation season; fit w^* on past data
- 2: Build the round graph and compute $\hat{v}_t$ using Eq. (3)
- 3: Predict $\hat{e}_t = \arg \min_i [\alpha_t q_{t,i} + (1 - \alpha_t) \hat{v}_{t,i}]$
- 4: **After observing E_t :** construct all constraints in Eq. (6)
- 5: Solve Eq. (7); record feasibility and solver status
- 6: Compute feasible-set geometry and conditional diagnostics

The temperature τ provides a smooth surrogate for deterministic elimination, and n_t rescales simplex-valued scores appropriately. This is a working likelihood, not a claim that real voting is literally a softmax sampler. Modeling the expert contribution removes one source of ambiguity present in a survival-only objective, but it does not, by itself, resolve the underlying inverse problem.

3.2 Outcome-Conditioned Reconstruction

Once E_t is observed, every eliminated contestant e and surviving contestant s must satisfy $c_{t,e} \leq c_{t,s}$. Equation (1) turns this into an explicit linear inequality:

$$v_{t,e} - v_{t,s} \leq \frac{\alpha_t}{1 - \alpha_t} (q_{t,s} - q_{t,e}), \quad e \in E_t, s \in C_t \setminus E_t. \quad (6)$$

Stacking these inequalities gives $A_t v \leq b_t$, so the feasible set and reconstructed vote are

$$\mathcal{P}_t = \{v \in \Delta^{n_t-1} : A_t v \leq b_t\}, \quad v_t^* = \arg \min_{v \in \mathcal{P}_t} \frac{1}{2} \|v - \hat{v}_t\|_2^2. \quad (7)$$

Whenever $\mathcal{P}_t$ is nonempty, this Euclidean projection is unique. All eliminated-survivor pairs are included, even in the general multiple-elimination formulation; the prediction benchmark we implement nevertheless restricts to single-elimination rounds, so it never obtains the number of eliminations from the test labels.

Because the inequality is weak, ties are permitted. We therefore report *tie-compatible feasibility*, using a numerical tolerance of 10^{-7} , rather than claiming that every arbitrary tie-breaking rule must reproduce E_t . A solver failure is recorded as a failure and never silently replaced by a uniform vote treated as evidence. Rank-based rules generally induce a union of order regions rather than a single percentage-rule polytope, and judge-save rounds additionally require modeling an intermediate bottom-two event; we exclude both from the present reconstruction benchmark.

3.3 Uncertainty in the Affine Simplex

An anchor picks out a single feasible distribution, but the relevant uncertainty concerns the entire feasible set. We report coordinate ranges $[\min_{v \in \mathcal{P}_t} v_i, \max_{v \in \mathcal{P}_t} v_i]$ obtained from linear programs, together with an intrinsic Chebyshev radius.

Let $\bar{A}_t v \leq \bar{b}_t$ collect both Eq. (6) and the nonnegativity facets $-Iv \leq 0$. With $P = I - \mathbf{1}\mathbf{1}^\top/n_t$, the largest ball contained in the simplex’s affine hull solves

$$\max_{z \in \mathbb{R}^{n_t}, r \geq 0} r \quad \text{s.t.} \quad \bar{a}_k^\top z + \|P\bar{a}_k\|_2 r \leq \bar{b}_k \quad \forall k, \quad \mathbf{1}^\top z = 1. \quad (8)$$

Only r is explicitly constrained to be nonnegative; nonnegativity of z follows automatically from the facet inequalities. We normalize as $R_t = r_t \sqrt{n_t(n_t - 1)} \in [0, 1]$, using the inradius of the full simplex as reference. Omitting the nonnegativity facets, or treating an optimization failure as a radius of one, would invalidate this interpretation. It is also worth stressing that radius is not volume, and that neither the radius nor a coordinate range is a calibrated posterior probability.

3.4 Topology and Counterfactual Diagnostics

We deliberately distinguish a flow residual from persistent homology. On the complete comparison graph, define the antisymmetric contrast

$$F_{t,ij} = \tanh(3n_t(q_{t,i} - q_{t,j})) - \tanh(3n_t(v_{t,i}^* - v_{t,j}^*)). \quad (9)$$

The best additive potential s minimizes $\sum_{i < j} (F_{t,ij} - s_i + s_j)^2$ subject to $\sum_i s_i = 0$; for unit comparison weights this reduces to $s_i = n_t^{-1} \sum_j F_{t,ij}$. We report

$$\Gamma_t = \frac{\|F_t - (s\mathbf{1}^\top - \mathbf{1}s^\top)\|_F}{\|F_t\|_F + 10^{-12}}. \quad (10)$$

This quantity measures the non-additivity of the chosen nonlinear contrast [13]. Because even two individually well-ordered score vectors can produce a nonzero residual after a nonlinear transformation, Γ_t is not by itself evidence that voters hold intransitive preferences; a raw difference of scalar rank gaps, by contrast, would be an additive flow with zero residual.

For persistent homology, we treat the points in each round as $(n_t q_{t,i}, n_t v_{t,i}^*)$ for $i \in C_t$, using Euclidean distances, and compute Vietoris–Rips H_0 and H_1 over $\mathbb{F}_2$, including triangles. An interval (b, d) has lifetime $d - b$, and N_η counts intervals whose lifetime exceeds a stated threshold η ; this lifetime count should not be confused with the Betti number at a particular filtration scale. No absent contestants are padded with zero trajectories, and we validate the pipeline on a square, on collinear points, and on the agreement case $v^* = q$ before interpreting its outputs on real rounds.

Our counterfactual mechanisms are evaluated at fixed preferences under explicit, controlled shocks. Expert alignment is the Spearman correlation with q , and crowd retention is the correlation with the unshocked ground-truth votes in the synthetic experiment; because these two metrics can trade off against each other, we include an expert-only comparator, which attains perfect expert alignment by construction, as well as a fixed expert weight of 0.8, rather than treating expert alignment alone as a proxy for fairness.

Table 2. Evaluation scope. The archive is broader than the controlled chronological benchmark.

Quantity	Count
Seasons in the archived competition table	34
Contestant–season entries / distinct contestants	421 / 408
Observed contestant–week rows / rounds	2777 / 335
Eligible single-elimination percentage-rule rounds	170
Chronologically evaluated rounds / seasons	59 / 9
Synthetic test rounds per rule / independent seeds	1,000 / 5

4 Experiments

4.1 Data and Reproducible Protocol

We use the COMAP-format competition CSV from the TIPL artifact [14] at commit 50909da8ebb7; its hash, data snapshot, and exclusion list accompany the manuscript. The records contain expert scores and outcomes, but not crowd votes. Dated follower counts are excluded from prospective features.

We restrict the benchmark to percentage-rule seasons 3–27 and rounds with one elimination, at least three active contestants, and no withdrawal. Positive judge scores define active records and weekly expert shares; elimination labels come from the textual result. A score–label mismatch excludes season-18 weeks 1–2. This yields 170 eligible rounds and 59 tests across seasons 18–27; season 26 has no eligible round. The benchmark is therefore not an all-season evaluation.

The seven features and graph are defined in Section 2. Standardization and partner histories use training and strictly earlier-season data only; follower counts, final placement, and the current elimination label are excluded. For each test season, the latest earlier eligible season selects $\lambda \in \{0.001, 0.01, 0.1\}$, after which the model is refitted on all earlier data. All ablations are retrained, baselines use the same chronological records, and 95% intervals use paired resampling of the nine test seasons.

Utility models use L-BFGS-B with $\tau = 0.2$; logistic regression and gradient boosting use fixed hyperparameters. HiGHS and SLSQP compute feasibility, projections, and geometric summaries, with failures or constraint violations recorded explicitly. Complete feature constants, optimizer tolerances, fold assignments, and comparator settings are provided in the reproduction files.

4.2 Recovery with Known Synthetic Votes

Synthetic rounds let us access ground truth that is unavailable in the real competition data. Each round has $n \sim \text{Unif}\{6, \dots, 10\}$ contestants and six independent standard-normal covariates, generated as

$$q = \text{softmax}(0.9X_{\cdot 1} + 0.3\xi), \quad v^{\text{true}} = \text{softmax}(Xw_0 + 0.2\epsilon), \quad (11)$$

with $w_0 = (0.2, 0.9, -0.65, 0.35, 0, 0.15)^\top$ and independent standard-normal noise ξ, ϵ . The observed elimination is $\arg \min_i [\alpha q_i + (1 - \alpha)v_i^{\text{true}}]$. For each of five

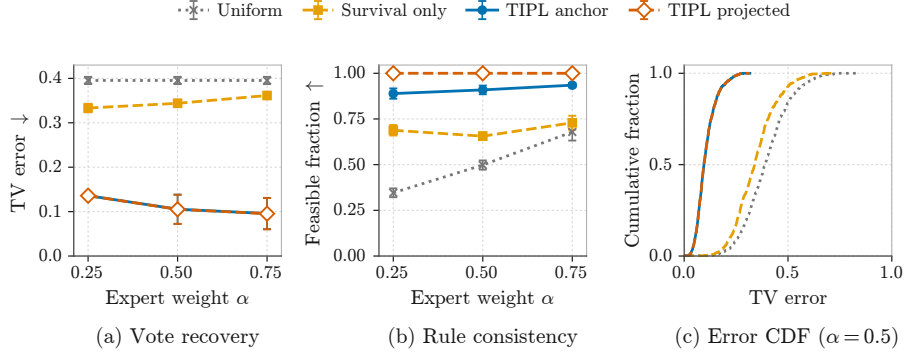


Fig. 2. Recovery and consistency answer different questions. (a,b) Mean and one standard deviation over five seeds, each with 200 test rounds per expert weight. (c) Empirical TV-error distributions on all 1,000 test rounds at $\alpha = 0.5$. Projection enforces feasibility with little change in recovery error, since it makes use of the observed outcome.

Table 3. Synthetic validation at $\alpha = 0.5$. TV is mean $\pm$ seed standard deviation; rank is mean Spearman correlation with the true votes. Uniform shares use a zero-correlation convention because all ranks tie.

Method	TV $\downarrow$	Rank $\uparrow$	Feasible (%) $\uparrow$
Uniform	0.395 ± 0.008	0.000	49.8
Survival-only anchor	0.344 ± 0.008	0.653	65.6
TIPL anchor	0.105 ± 0.033	0.950	90.9
TIPL projected	0.105 ± 0.033	0.951	100.0

seeds and each $\alpha \in \{0.25, 0.50, 0.75\}$, we generate 500 training, 150 validation, and 200 test rounds, withholding votes from both fitting and hyperparameter selection. This experiment is designed to isolate the inverse objective; it does not validate the competition-specific graph or social features.

We compare four approaches: uniform shares, a survival-only softmax anchor, our rule-aware anchor, and its outcome-conditioned projection. The survival-only objective maximizes the average log vote assigned to observed survivors, using the same covariates and regularization candidates, and both objectives select λ by validation elimination accuracy. We measure error as $\text{TV}(v, v^{\text{true}}) = \frac{1}{2} \|v - v^{\text{true}}\|_1$, and error bars denote the standard deviation across seeds rather than uncertainty about unobserved historical votes.

As Table 3 and Figure 2 show, the rule-aware anchor reduces TV from 0.344 to 0.105 in this matched generative setting, a substantial improvement. Projection, by contrast, changes TV only from 0.10539 to 0.10519, even though it raises the feasible fraction from 90.9% to 100%. The large gain in feasibility is therefore not evidence of a comparably large gain in vote recovery. We observe similar qualitative behavior at the other expert weights (Section 4.2). An intentionally misspecified reconstruction weight also changes feasibility and can even render the observed event impossible; we count all such solver failures rather than concealing

them. Together, these results support the value of the rule-aware observation model while also delimiting exactly what the projection step does and does not establish.

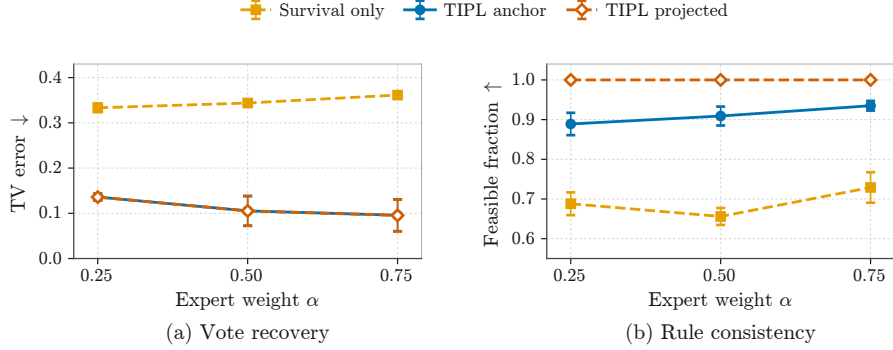


Fig. 3. Sensitivity to the expert mixing weight. Each point summarizes five independent seeds, each with 200 synthetic test rounds; error bars are standard deviations of seed means. The true rule is supplied to training and the correctly specified projection. Ground-truth votes are used only for evaluation.

Rule Sensitivity The synthetic generator uses seeds 2026–2030 independently for each true expert weight. Each seed generates training, validation, and test samples consecutively. The true coefficient vector and noise levels are given in Eq. (11). The same generated rounds are reused across methods. Models are fitted on the 500 training rounds; validation selects regularization and is not combined with training in this controlled experiment. The real-data experiment instead refits after selection, as specified above. This difference is explicit in the reproduction scripts.

To test rule misspecification, we project the learned anchor using an assumed expert weight of 0.8 regardless of the true weight. Table 4 counts failed solves in the denominator of true-rule feasibility. TV is reported only for successful solves, with success coverage alongside it. This prevents infeasible cases from silently disappearing from the comparison. The wrong rule can force an erroneous correction even when the original learned anchor is informative.

4.3 Prospective Elimination Prediction

All prediction methods are evaluated on exactly the same 59 rounds. Judge-only selects the contestant with the lowest expert share; logistic regression and gradient boosting instead learn elimination risk directly. The utility-based methods first produce an outcome-blind vote anchor and then apply the percentage rule to

Table 4. Misspecified projection with assumed $\alpha = 0.8$. Each row covers 1,000 test rounds. Feasibility is evaluated under the true rule. Conditional TV should be read together with solver coverage.

True α	Solver success (%)	TV (successful)	True-rule feasible (%)
0.25	94.7	0.161	90.4
0.50	99.7	0.112	95.2
0.75	100.0	0.095	97.1

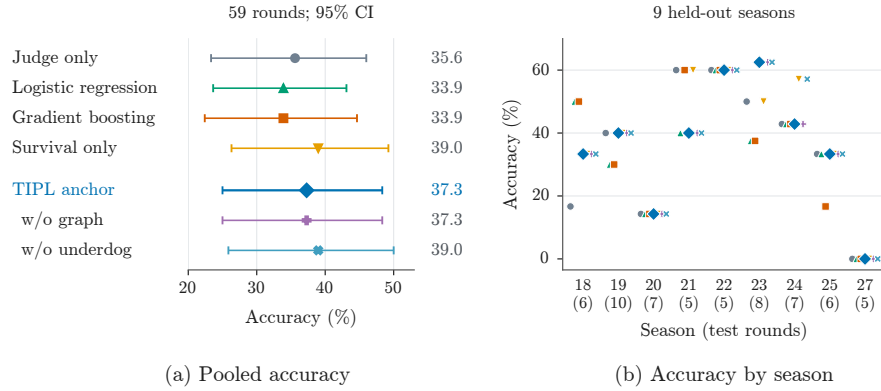


Fig. 4. Chronological prediction and retrained ablations. (a) Accuracy on 59 rounds with 95% intervals from 2,000 paired season-block resamples. LR denotes logistic regression; GBDT denotes gradient boosting. (b) Within-season accuracy, rounded to integer percentages; column labels show season and test-round count. Aggregate ablation results appear in Table 5.

it. Graph and underdog ablations are retrained with the corresponding features removed, and no projected votes are used anywhere in this prediction comparison.

The full anchor, the graph ablation, and the underdog ablation correctly predict 22, 22, and 23 rounds, respectively. These data do not establish either feature group’s necessity. Retrospective projection is deliberately absent from this table because it uses the answer to the prediction task. Its 100% feasibility is reported separately as a property of satisfying a constructed constraint set. Weak inequalities also mean that feasibility and a particular deterministic tie-breaking outcome need not coincide.

TIPL correctly predicts 22/59 rounds (37.29%), compared with 21/59 for judge-only (35.59%) and 23/59 for the survival-only anchor (38.98%). The graph ablation also predicts 22/59 rounds correctly, while removing the underdog feature yields 23/59. These point estimates do not display the strong advantage that retrospective consistency alone would suggest, and with overlapping uncertainty intervals across only nine season blocks, we do not treat this as a formal

Table 5. Outcome-blind prediction and retrained ablations. All methods use the same 59 test rounds. Intervals are season-block bootstrap intervals; they are descriptive uncertainty summaries based on nine blocks.

Method	Accuracy (%)	95% interval
Judge only	35.59	[23.3, 46.0]
Logistic regression	33.90	[23.6, 43.1]
Gradient boosting	33.90	[22.4, 44.6]
Survival-only anchor	38.98	[26.3, 49.2]
TIPL anchor	37.29	[25.0, 48.3]
TIPL without graph	37.29	[25.0, 48.3]
TIPL without underdog	38.98	[25.9, 50.0]

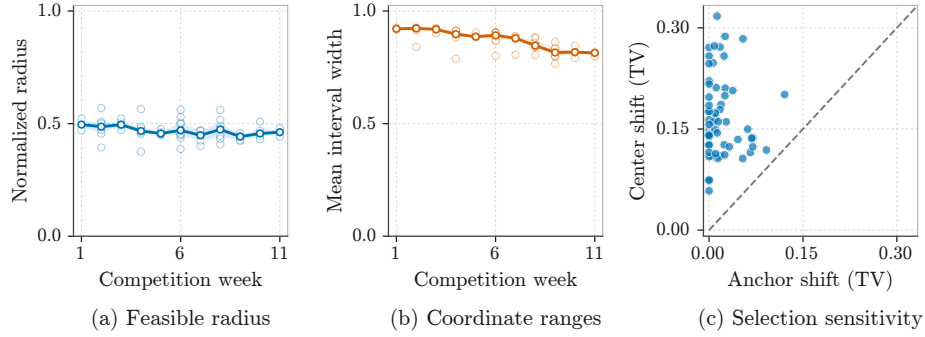


Fig. 5. Observed eliminations leave broad feasible vote sets. (a,b) Individual rounds, weekly medians, and interquartile bands. Radius is normalized by the simplex inradius; widths average the feasible coordinate intervals. (c) Each round’s TV distance from the projection to the anchor and to the feasible-set center, with the dashed line marking equality. These are geometric ranges, not posterior credible intervals.

equivalence result; the predictive evidence remains limited. Figure 4 is intended to convey this uncertainty rather than to assert state-of-the-art accuracy.

4.4 Retrospective Geometry and Structural Diagnostics

Conditioning on the observed test-round elimination renders all 59 projected distributions feasible by construction; this is simply the expected consequence of solving Eq. (7), not an independent accuracy result. The mean TV change from anchor to projection is 0.020, while the mean TV distance between the projection and the feasible-set center is 0.168. In other words, different reasonable selection criteria can yield substantially different votes even when conditioned on the same observed outcome.

The median normalized radius is 0.468, and the mean coordinate-range width is 0.871. Figure 5 is inconsistent with treating late-round outcomes as uniquely pinning down crowd votes: coordinate endpoints are optimized separately and need not be simultaneously attainable. This broad range is a direct consequence of partial identification, and it should accompany any behavioral interpretation of an individual reconstruction.

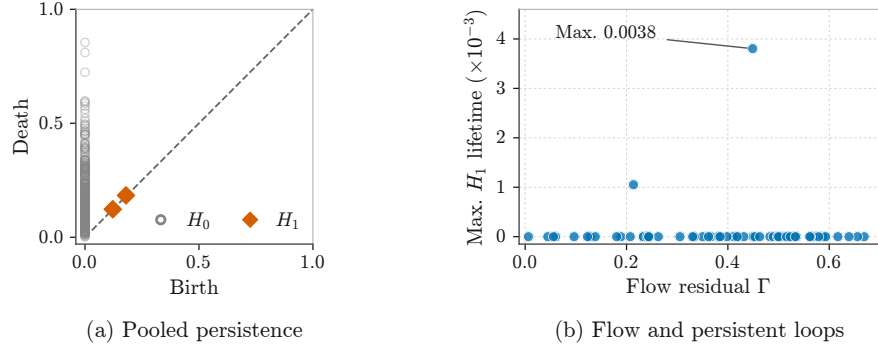


Fig. 6. Flow non-additivity does not establish persistent loops. (a) All finite intervals pooled from the 59 round-level point clouds; circles show H_0 and diamonds H_1 . (b) Flow residual versus the longest H_1 lifetime in each round, including zeros. The lifetime axis uses a 10^{-3} scale; no round exceeds the tested threshold 0.05.

The mean flow residual is $\Gamma = 0.381$, while the largest computed H_1 lifetime is only 0.0038; no interval exceeds 0.05, 0.10, or 0.15 in any of these rounds. Figure 6, together with our agreement-case control, therefore does not support an inference of persistent cyclic or multi-polar crowd conflict. A nonlinear flow residual and a topological loop are conceptually different objects, and neither one can identify the psychological cause of disagreement from censored votes alone.

Topological Controls and Interpretation The persistent-homology implementation sorts vertices, edges, and triangles by filtration value, breaking equal-value ties by dimension and vertex index. Boundary columns are reduced over $\mathbb{F}_2$. Finite intervals with positive lifetime are saved; the infinite connected component is omitted from persistence plots. On the four corners of a unit square, it returns the expected H_1 interval $[1, \sqrt{2})$. Collinear points have no positive-lifetime H_1 interval. The expert-crowd agreement control gives zero contrast flow and a zero residual. The real-data persistence analysis has no interval above any of the tested thresholds 0.05, 0.10, or 0.15.

The point-cloud definition is per round. It avoids interpreting correlations between zero-padded post-elimination trajectories as observed relationships. The flow analysis likewise names the contrast being decomposed; it does not identify a Hodge residual with persistent homology. All intervals in the reported diagrams are computed from the specified cloud and saved with their round identifiers.

4.5 Counterfactual Stress Tests

We use the first 100 synthetic test rounds from each seed at $\alpha = 0.5$ to obtain 500 paired cases with known votes. A popularity shock reallocates mass toward the contestant with the lowest expert score:

$$v^{\text{shock}} = (1 - \zeta)v^{\text{true}} + \zeta \mathbf{e}_{i_{\min q}}, \quad \zeta \in \{0, 0.25, 0.50, 0.75\}. \quad (12)$$

We compare this against equal percentage aggregation, equal rank aggregation, a fixed expert weight of 0.8, and expert-only scoring. As an illustrative adaptive

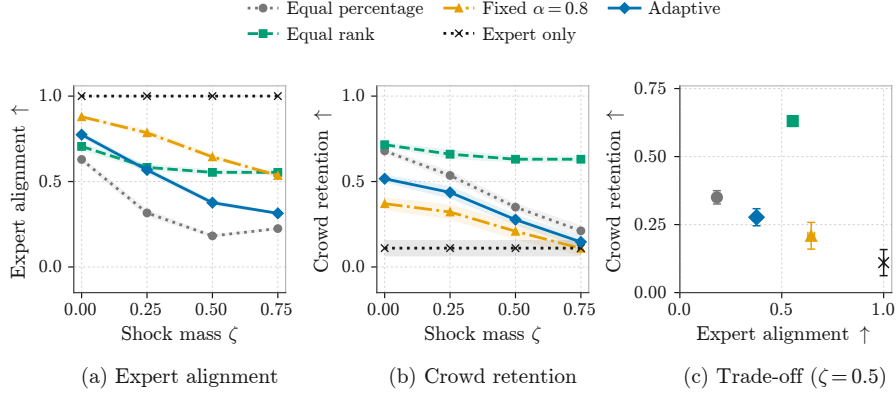


Fig. 7. Expert alignment has a crowd-retention cost. (a,b) Means and one-standard-deviation bands over five seeds, each with 100 paired synthetic rounds. (c) The two metrics at shock mass 0.5, with one-standard-deviation error bars. Expert-only alignment is one by construction; the adaptive policy does not dominate the alternatives on both metrics.

policy, we also test

$$\alpha_{\text{adapt}} = \text{clip}_{[0.5, 0.9]} (0.5 + 0.3\Gamma(q, v^{\text{shock}}) + 0.1R), \quad (13)$$

where R is computed from the un-shocked observed event and its coefficients are fixed rather than tuned on stress-test outcomes. We emphasize that this is a retrospective simulation under an assumed preference environment, not a deployed policy or a strategic equilibrium.

At shock mass 0.5, adaptive aggregation achieves expert alignment 0.376 and crowd retention 0.277, compared with 0.182 and 0.350 for equal percentages. Fixed expert weight 0.8 achieves 0.644 and 0.209. These results expose a genuine trade-off rather than demonstrating universal robustness: they do not, on their own, justify a fairness score without an accompanying normative definition, nor do they show that adaptive aggregation is optimal. Throughout, these tests hold voters’ baseline preferences fixed and do not attempt to model how those preferences might adapt to a changed mechanism.

Counterfactual Interpretation The stress test uses 500 known-preference synthetic rounds, fixed before applying shocks. Spearman correlations are computed against the expert vector and the original crowd vector; a constant score vector is assigned correlation zero because it contains no ranking information. Five seed means determine the plotted standard-deviation bands. The point of the expert-only comparator is to expose the metric’s built-in objective: a score with perfect expert alignment need not retain crowd preference. No separate fairness quantity is reported because the earlier draft did not operationalize it sufficiently.

5 Discussion and Limitations

What the reconstruction means. TIPL selects one model-dependent point from a feasible set. Synthetic recovery holds only under the specified generator and noise model; real crowd votes remain unobserved. Thus neither fitted coefficients, retrospective agreement, nor inclusion of known expert scores identifies the true magnitude of popularity, sympathy, or partner effects.

What the evaluation covers. The archive spans 34 seasons, but the controlled benchmark covers nine test seasons and restricted rounds. Rule changes, incomplete documentation, and missing contemporaneous popularity data limit external validity, and graph features show no predictive advantage. Broader validation requires timestamped features, observed vote shares or repeated preference measurements, and explicit models of rank-based and discretionary-save rules.

What the diagnostics establish. Geometry measures how under-determined the feasible set is. The nonlinear flow statistic measures non-additivity, but neither diagnostic warrants a causal interpretation. Persistent homology is exploratory, depends on the chosen point cloud and scale, and yields a negative result that we report unchanged. Expert alignment is not fairness. Further work across additional domains should test alternative preference generators, calibrated uncertainty, and models of strategic responses before recommending any voting rule.

6 Conclusion

TIPL combines a rule-aware utility anchor, an explicit feasible-set reconstruction, and structural diagnostics for auditing hybrid voting systems. Our evaluation carefully distinguishes a predictive test from an outcome-conditioned explanation, and validates vote recovery only in the setting where ground truth is actually available. Projection enforces the constraints implied by an observed outcome, while the remaining feasible set quantifies just how much that outcome leaves unresolved. Together, these pieces support transparent, assumption-dependent mechanism auditing, with clearly stated limits on what can be claimed about true preferences and about robustness.

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